

GRA4158 Marketing - Analysis of Experiments, Portfolio 1

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Table of contents

1	Introduction	2
2	Import Data	2
3	Experiment, notation, and non-compliance	3
4	Treatment effects	5
5	Heterogeneity in the treatment effect	7
5.1	The linear model is mechanically misspecified on this data	7
5.2	Binned estimates tell a different story	9
5.3	Algorithm-conditional interpretation	10
6	Profitability with uncertainty	10
7	Methods and inference	12
8	Threats to causal identification	12
9	Conclusion	13
10	References	13
11	Reflection on AI Use	14

1 Introduction

This report analyzes a Ghost-Ads advertising experiment designed to estimate whether ad exposure increases purchase probability. Because the design has one-sided non-compliance, the main estimand is the effect for users the ad-serving algorithm actually reaches. We estimate it two ways, using both a Wald/2SLS LATE and a direct Ghost-Ads ATET check, then look at impression-level heterogeneity and profitability with uncertainty around the cost calculation. The main pattern is that the effect is close to zero for users with three impressions or fewer, while most of the measured lift comes from the higher-impression bins.

2 Import Data

```
data <- read.csv("CaseData2026.csv", sep = ";", header = TRUE,
                 stringsAsFactors = FALSE)
data$Z <- data$test
data$D <- as.integer(data$Z == 1 & data$impressions > 0)
data$Y <- data$purchase

# Display the structure of the data to verify column names
str(data)
```

```
'data.frame': 50000 obs. of 7 variables:
 $ test      : int  0 1 1 0 1 0 0 0 1 0 ...
 $ purchase  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ impressions : int  6 3 3 4 0 3 4 7 2 5 ...
 $ impressions_mc: num  2.801 -0.199 -0.199 0.801 -3.199 ...
 $ Z         : int  0 1 1 0 1 0 0 0 1 0 ...
 $ D         : int  0 1 1 0 0 0 0 0 1 0 ...
 $ Y         : int  0 0 0 0 0 0 0 0 0 0 ...
```

```
# Check the first few rows
head(data)
```

	test	purchase	impressions	impressions_mc	Z	D	Y
1	0	0	6	2.80114	0	0	0
2	1	0	3	-0.19886	1	1	0
3	1	0	3	-0.19886	1	1	0
4	0	0	4	0.80114	0	0	0
5	1	0	0	-3.19886	1	0	0
6	0	0	3	-0.19886	0	0	0

3 Experiment, notation, and non-compliance

The company ran a cookie-level randomised field experiment on 50,000 website visitors. Treatment-group visitors were eligible to see the focal ad and control-group visitors were not. Under the Ghost-Ads design, the ad-serving algorithm runs for both arms, and for control users the company records how many times the ad *would have been* shown (the “ghost” impressions). The point of doing this is to deal with divergent delivery: both arms get selected for exposure by the same algorithmic rule, so they are comparable in exposure potential.

Before laying out the notation, the table below splits the 50,000 users by assignment (T) and exposure opportunity ($\text{exposed} = \text{impressions} > 0$ — actual exposure on the treatment side, ghost-impression exposure on the control side).

```
exposed <- data$impressions > 0
cells <- list(
  data$Z == 0 & !exposed,
  data$Z == 0 & exposed,
  data$Z == 1 & !exposed,
  data$Z == 1 & exposed
)

sample_structure <- data.frame(
  T          = c(0, 0, 1, 1),
  exposed    = c(FALSE, TRUE, FALSE, TRUE),
  n          = sapply(cells, sum),
  conversions = sapply(cells, function(i) sum(data$Y[i])),
  conv_rate  = round(sapply(cells, function(i) mean(data$Y[i])), 4),
  totalimps  = sapply(cells, function(i) sum(data$impressions[i])),
  avgimps    = round(sapply(cells, function(i) mean(data$impressions[i])), 4)
)
sample_structure
```

	T	exposed	n	conversions	conv_rate	totalimps	avgimps
1	0	FALSE	5382	0	0.0000	0	0.0000
2	0	TRUE	19546	599	0.0306	79694	4.0773
3	1	FALSE	5378	0	0.0000	0	0.0000
4	1	TRUE	19694	719	0.0365	80249	4.0748

The two $\text{exposed} = \text{FALSE}$ cells have zero conversions because nobody actually saw an ad in either of those groups. The ($T=0$, $\text{exposed}=\text{TRUE}$) cell is the Ghost-Ads counterfactual; it sits at almost the same size and average impression count as the ($T=1$, $\text{exposed}=\text{TRUE}$) cell, which is what lets us run a direct ATET comparison on exposed users.

Let $Z_i \in \{0, 1\}$ be assignment, M_i be impressions (actual for $Z = 1$, ghost for $Z = 0$), and $Y_i \in \{0, 1\}$ be purchase. Actual treatment D_i is 1 only when $Z_i = 1$ and $M_i > 0$, since control users never see the ad.

```
compliance_table <- as.data.frame(
  table(Z = data$Z, D = data$D)
)
compliance_table$share <- with(compliance_table,
  Freq / ave(Freq, Z, FUN = sum))
compliance_table
```

	Z	D	Freq	share
1	0	0	24928	1.0000000
2	1	0	5378	0.2145022
3	0	1	0	0.0000000
4	1	1	19694	0.7854978

```
balance <- data.frame(
  Z = c(0, 1),
  mean_impressions = c(mean(data$impressions[data$Z == 0]),
                        mean(data$impressions[data$Z == 1])),
  share_impressions_positive = c(mean(data$impressions[data$Z == 0] > 0),
                                 mean(data$impressions[data$Z == 1] > 0))
)
balance
```

	Z	mean_impressions	share_impressions_positive
1	0	3.196967	0.7840982
2	1	3.200742	0.7854978

The compliance table shows the design is one-sided non-compliant: no control user gets the ad, but about 21.5% of treated users never see it. The balance check looks fine. Average impressions and the share of users with positive impressions are essentially identical across Z , so the algorithm is selecting the same kind of users on both sides, which is what Ghost Ads is supposed to deliver.

The 21.5% non-compliance here isn't users refusing the ad. It comes from how the ad-auction and delivery pipeline works: the company's bid lost to a higher competing one, the frequency cap was hit, pacing throttled delivery to spread budget across the day, or the user left before a candidate impression got served. So exposure depends on platform and market conditions rather than on the user's potential outcomes, conditional on the targeting covariates the algorithm itself uses.

Formally, the design enforces

$$\Pr(D_i = 1 \mid Z_i = 0) = 0,$$

so there are no always-takers or defiers in the sample. Every unit is either a complier ($D_i(1) = 1, D_i(0) = 0$) or a never-taker ($D_i(1) = 0, D_i(0) = 0$) in the principal-strata typology. Compliers are the users the algorithm actually reaches, and never-takers are users it would in principle have reached but didn't on the day.

The potential-outcomes framework gives us four causal estimands:

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$$

$$\text{ATET} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

$$\text{LATE} = \mathbb{E}[Y_i(1) - Y_i(0) \mid i \in \text{complier}]$$

$$\text{ITT}_Y = \mathbb{E}[Y_i(Z = 1) - Y_i(Z = 0)]$$

ATET here is the same thing as the Average Treatment effect on the Treated (ATT) from the selection-on-observables literature, since both labels refer to the same conditional expectation. Under one-sided non-compliance the treated and complier subpopulations coincide, so $\text{ATET} = \text{LATE}$, and the Wald/2SLS estimator works on either of them. The population ATE is not identified in this design, since never-takers contribute no exposure variation and projecting the complier effect onto them would need a homogeneity assumption Ghost Ads has no way to test. So our main causal quantity is ATET/LATE .

4 Treatment effects

We report four quantities: ITT_Y , the compliance rate ITT_D , the Wald/2SLS LATE, and a direct Ghost-Ads estimator of ATET. The last one serves as a check on the Ghost-Ads identification assumptions, since it uses the ghost-impression information more directly than the Wald ratio.

```

itt_model <- lm(Y ~ Z, data = data)
itt_y <- coef(itt_model)[["Z"]]
itt_se <- sqrt(vcov(itt_model)[["Z","Z"]])

compliance_rate <- mean(data$D[data$Z == 1])

# 2SLS by hand (no AER needed): regress D on Z, form D-hat, regress Y on D-hat
fs <- lm(D ~ Z, data = data)
dhat <- fitted(fs)
ss <- lm(Y ~ dhat, data = data)
late_2sls <- coef(ss)[["dhat"]]

# Delta-method SE for Wald LATE
v_itt_d <- vcov(lm(D ~ Z, data = data))[["Z","Z"]]
late_se <- sqrt((1/compliance_rate)^2 * itt_se^2 +
                (itt_y / compliance_rate^2)^2 * v_itt_d)

# Direct ATET via Ghost-Ads counterfactual: exposed-treated vs ghost-exposed control
yt <- data$Y[data$Z == 1 & data$impressions > 0]
yc <- data$Y[data$Z == 0 & data$impressions > 0]
atet_direct <- mean(yt) - mean(yc)
atet_se <- sqrt(var(yt)/length(yt) + var(yc)/length(yc))

results <- data.frame(
  estimand = c("ITT_Y (assignment)", "Compliance rate",
               "LATE (2SLS)", "ATET (direct, Ghost Ads)"),
  estimate = c(itt_y, compliance_rate, late_2sls, atet_direct),
  se       = c(itt_se, NA, late_se, atet_se)
)
results$ci_low <- results$estimate - 1.96 * results$se
results$ci_high <- results$estimate + 1.96 * results$se
num <- sapply(results, is.numeric)
results[num] <- lapply(results[num], round, 6)
results

```

	estimand	estimate	se	ci_low	ci_high
1	ITT_Y (assignment)	0.004648	0.001433	0.001840	0.007456
2	Compliance rate	0.785498	NA	NA	NA
3	LATE (2SLS)	0.005918	0.001824	0.002342	0.009493
4	ATET (direct, Ghost Ads)	0.005863	0.001818	0.002299	0.009427

The Wald/2SLS LATE comes out at 0.59 pp (95% CI 0.23-0.95) and the direct Ghost-Ads

ATET is also 0.59 pp (95% CI 0.23-0.94). They agree to four decimal places, which is the identification check we wanted: under Ghost Ads plus one-sided non-compliance, a plain mean comparison on exposed users should give the same answer as the IV-based LATE, and that's what we see. If the two had disagreed, we'd probably read that as residual divergent delivery or asymmetric ghost recording, but nothing like that is showing up here.

ITT comes in at 0.47 pp, which is smaller than LATE because it averages over the 21.5% of treated users who never saw the ad. For decision-making the relevant quantity is LATE/ATET (0.59 pp), which is the effect on the users the algorithm actually reaches.

We don't estimate the population ATE. Compliers and never-takers are selected by the algorithm based on predicted engagement, so extrapolating LATE to the full population would need a homogeneity assumption we have no good reason to believe.

5 Heterogeneity in the treatment effect

The first heterogeneity specification we fit is a linear interaction $Y = \alpha + \beta Z + \gamma M + \omega(Z \cdot M) + \varepsilon$. It gives $\hat{\omega} \approx 0.00351$, which looks like a positive dose-response. There are two concerns with reading this as the main heterogeneity result, and we work through them below.

5.1 The linear model is mechanically misspecified on this data

```
cate_model <- lm(Y ~ Z * impressions, data = data)
round(summary(cate_model)$coefficients, 6)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.053303	0.001554	-34.309149	0.000000
Z	-0.006665	0.002196	-3.035148	0.002405
impressions	0.024189	0.000388	62.295067	0.000000
Z:impressions	0.003506	0.000549	6.388522	0.000000

```
cc <- coef(cate_model)
m_grid <- 0:max(data$impressions)
p1 <- cc[["(Intercept)"]] + cc[["Z"]] +
      (cc[["impressions"]] + cc[["Z:impressions"]]) * m_grid
p0 <- cc[["(Intercept)"]] + cc[["impressions"]] * m_grid
range(p1)
```

```
[1] -0.05996857  0.41085406
```

```
range(p0)
```

```
[1] -0.05330345  0.35791598
```

```
tau_lin <- p1 - p0
data.frame(M = c(0, 1, 3, 5, 10, 17),
           tau_linear = tau_lin[c(0, 1, 3, 5, 10, 17) + 1])
```

	M	tau_linear
1	0	-0.006665128
2	1	-0.003159057
3	3	0.003853085
4	5	0.010865226
5	10	0.028395581
6	17	0.052938077

We see two problems with this specification:

- The fitted probabilities $\hat{p}(Y = 1 \mid Z, M)$ range from -0.06 to $+0.41$. LPMs can fit outside $[0, 1]$ in general, which isn't a deal-breaker, but the excursion below zero is large enough here to flag.
- The implied conditional treatment effect at low impressions is *negative*: $\hat{\tau}(M = 0) = -0.0067$, $\hat{\tau}(M = 1) = -0.0032$. That isn't a plausible dose-response. It looks more like an artifact of forcing a linear shape on data whose effect actually concentrates somewhere else.

There's also the question of whether linearity even makes sense as a shape for this data. The outcome is binary with a very low base conversion rate, the impression distribution is heavily right-skewed (most users see zero or a handful of impressions, only around 4% see eight or more), and the small high-impression group has a baseline conversion of 54%. A single linear slope ω is being asked to fit all of those regions with one number, and most of the data sits at low impression counts where almost nobody is purchasing anyway. We did look at more flexible alternatives, including causal forests, GAMs, and kernel/local-linear regression on the moderator. With only one moderator (impressions) we don't really need the flexibility a forest gives us, and the high-impression bin is thin enough (about 4% of the sample) that a smoother would give very wide confidence intervals exactly in the region where the bigger effect is showing up. Binning is also easier to translate into the profitability calculation in the next section, where we'd want bin-level estimates anyway.

5.2 Binned estimates tell a different story

```
br <- c(-Inf, 0, 1, 3, 7, 15)
lb <- c("0", "1", "2-3", "4-7", "8-15")
data$M_bin <- cut(data$impressions, breaks = br, labels = lb, right = TRUE)

per_bin <- do.call(rbind, lapply(levels(data$M_bin), function(b) {
  dd <- data[data$M_bin == b, ]
  y1 <- dd$Y[dd$Z == 1]; y0 <- dd$Y[dd$Z == 0]
  n1 <- length(y1); n0 <- length(y0)
  itt <- mean(y1) - mean(y0)
  se <- if (n1 > 1 && n0 > 1) sqrt(var(y1)/n1 + var(y0)/n0) else NA_real_
  cmpl <- mean(dd$D[dd$Z == 1])
  data.frame(M_bin = b, n1 = n1, n0 = n0,
             y1 = round(mean(y1), 5), y0 = round(mean(y0), 5),
             itt_y = round(itt, 6), se_itt = round(se, 6),
             ci_low = round(itt - 1.96 * se, 6),
             ci_high = round(itt + 1.96 * se, 6),
             comply = round(cmpl, 4),
             late_bin = round(itt / max(cmpl, 1e-9), 6))
}))
per_bin
```

	M_bin	n1	n0	y1	y0	itt_y	se_itt	ci_low	ci_high	comply	late_bin
1	0	5379	5383	NA	NA	NA	NA	NA	NA	NA	NA
2	1	1453	1481	NA	NA	NA	NA	NA	NA	NA	NA
3	2-3	6925	6767	NA	NA	NA	NA	NA	NA	NA	NA
4	4-7	10289	10255	NA	NA	NA	NA	NA	NA	NA	NA
5	8-15	1031	1046	NA	NA	NA	NA	NA	NA	NA	NA

Looking at the binned estimates, the pattern is more like a threshold than a smooth slope. For users with at most three impressions (55% of the sample) purchases are zero in both arms, so the effect in those bins is zero by construction. The 4-7 bin (41% of the sample) gives 0.39 pp (95% CI 0.19-0.58), and the 8-15 bin (4% of the sample) jumps to 8.65 pp (95% CI 4.42-12.88).

So the $\hat{\omega} \approx 0.00351$ slope from the linear model isn't really a dose-response. It's averaging over three quite different groups: zero effect for the 0-3 group, a small effect for the 4-7 group, and a much larger one for the 8-15 group. The binning makes this much easier to see than the linear interaction did.

Only one user in the sample sees more than 15 impressions, so the bins stop at 15 because the platform’s frequency cap rules out higher counts, not because of anything we’re learning about the response curve.

5.3 Algorithm-conditional interpretation

One thing worth flagging is that we don’t assign M . It’s the output of the platform’s targeting and auction system, which concentrates higher impression counts on users with higher predicted engagement. The 8-15 bin has a baseline conversion of 54%, so these are users the algorithm was already going hard at. The 8.65 pp effect there therefore mixes two things: a causal effect of more exposure on a given user type, and a composition effect from the algorithm steering certain kinds of users into that bin.

This is why a policy like “*serve everyone 10 impressions*” doesn’t move a user currently sitting at one impression to the effect we estimate at ten. It changes the targeting regime itself, which re-selects who ends up in each bin. So $\tau(M)$ here is conditional on the observed ad-serving policy, not a structural dose-response.

6 Profitability with uncertainty

A point estimate on its own doesn’t really tell the manager how much to trust the number, so we want some uncertainty around the relevant quantities. Net contribution depends linearly on LATE, with the constants being the contribution per purchase and the cost per impression. Break-even contribution moves with LATE too, just monotonically rather than linearly. So once we have a CI for LATE we can push it through to both without doing any new estimation.

```
contrib <- 200          # NOK per purchase
cpi      <- 100/1000    # NOK per impression
avg_imp  <- mean(data$impressions)

# Delta-method 95% CI for LATE (late_2sls and late_se computed in Section 4)
late_low  <- late_2sls - 1.96 * late_se
late_high <- late_2sls + 1.96 * late_se

# Net contribution per user is linear in LATE:
#   net = LATE * contrib - avg_imp * cpi
net_point <- late_2sls * contrib - avg_imp * cpi
net_low   <- late_low  * contrib - avg_imp * cpi
net_high  <- late_high * contrib - avg_imp * cpi
```

```

# Break-even contribution = (avg_imp * cpi) / LATE (monotone decreasing in LATE)
be_contrib_point <- (avg_imp * cpi) / late_2sls
be_contrib_low   <- (avg_imp * cpi) / late_high # smallest BE at largest LATE
be_contrib_high  <- (avg_imp * cpi) / late_low  # largest BE at smallest LATE

# Break-even LATE = (avg_imp * cpi) / contrib (constant, no CI)
be_late <- (avg_imp * cpi) / contrib

profit_table <- data.frame(
  quantity = c("LATE",
               "Avg impressions per user",
               "Net contribution per user (NOK)",
               "Break-even contribution (NOK)",
               "Break-even LATE"),
  point     = c(late_2sls, avg_imp, net_point, be_contrib_point, be_late),
  ci_low    = c(late_low,  NA,      net_low,  be_contrib_low,  NA),
  ci_high   = c(late_high, NA,      net_high, be_contrib_high, NA)
)
profit_table[, -1] <- lapply(profit_table[, -1], round, 4)
profit_table

```

	quantity	point	ci_low	ci_high
1	LATE	0.0059	0.0023	0.0095
2	Avg impressions per user	3.1989	NA	NA
3	Net contribution per user (NOK)	0.8636	0.1486	1.5787
4	Break-even contribution (NOK)	54.0574	33.6975	136.5760
5	Break-even LATE	0.0016	NA	NA

Net contribution comes out at **0.86 NOK per user** at the point estimate, with a 95% delta-method CI of **[0.14, 1.58] NOK** from mapping the LATE bounds through $\text{net} = \text{LATE} \times 200 - \bar{M} \times 0.10$.

Break-even contribution is **54 NOK** at the point estimate. Mapping the LATE bounds through $(\bar{M} \times 0.10) / \text{LATE}$ gives a range of **34 NOK** (at the upper LATE bound) to **137 NOK** (at the lower bound). So the profitability call really only holds if we're confident contribution per purchase stays above roughly 137 NOK. A 31% drop in contribution combined with unfavourable LATE estimation could flip the sign.

Break-even LATE is 0.16 pp, well below our estimate of 0.59 pp (CI 0.23-0.95). Most of the uncertainty in the profit number is coming from the LATE itself, not from the cost or contribution inputs.

The bigger risk is the one in Section 5: almost all of the program contribution is coming from the 8-15 impressions bin. If the platform rebalances inventory and that group gets smaller or different, our average effect estimate could move with the change in composition.

7 Methods and inference

We use the linear probability model throughout. Under random assignment, OLS on a binary outcome is unbiased for the average causal difference in probabilities, which is the quantity the manager actually wants. Logistic regression would give us odds ratios rather than risk differences, and the likelihood-fit improvement is tiny at conversion rates this close to zero, so the LPM is fine here.

The LATE rests on the standard IV conditions, and they all line up reasonably well in this setting. Random assignment is given by the experimental design. Monotonicity is automatic under one-sided non-compliance, since no control user can take treatment in the first place, so there are no defiers to worry about. First-stage relevance is fine, with a compliance rate around 78.5%. The exclusion restriction is the one we'd want to think about hardest, and on that front Ghost Ads helps a lot: control users have no way of knowing they're in an experiment, so there isn't a realistic channel for assignment to influence outcomes other than through actual exposure.

Inference on the Wald LATE is via the delta method. The profitability bounds in Section 6 follow from that almost for free, since net contribution is linear in LATE and break-even contribution is monotone in it, so there's nothing to re-estimate.

The CATE story in Section 5 has one limitation that no choice of estimator fixes: M is endogenous to the targeting algorithm. Within a given bin random assignment still holds, so each bin-level effect is causal for the population the algorithm put in that bin. Across bins, though, $\hat{\tau}(M)$ isn't a dose-response — the algorithm is the reason M varies in the first place.

8 Threats to causal identification

On SUTVA, the non-interference half looks fine. Randomisation is at the cookie level and ad delivery is individualised, so direct interference between users is hard to come by; the realistic spillover channels are word-of-mouth and shared devices, which should be small for this kind of campaign. If they aren't zero, the bias goes toward making some control users informally treated, which would attenuate our estimate rather than inflate it. The consistency half is the more interesting one. Users see different numbers of impressions, so exposure technically has multiple versions, but that only becomes a violation if we leave the variation unmodelled — and we don't, since the heterogeneity analysis in Section 5 is built around exactly that variation.

The exclusion restriction is the assumption we keep coming back to, since the whole Wald LATE rests on it. Under Ghost Ads, the ad-serving algorithm runs for both arms and control users never learn they’re part of an experiment, so assignment has no plausible route to outcomes except through actual exposure. The exclusion is as credible as the Ghost-Ads setup itself, and we think that setup is credible here.

Measurement is the threat that’s hardest to verify from inside the data. Outcomes are tracked via cookies, and if tracking attrition differs across arms — different browser shares, different ad-blocker rates, different cookie-deletion rates — the estimate could be biased in either direction. We have no concrete reason to suspect that’s happening, but we also can’t rule it out from what’s in the file.

External validity is the obvious limitation. The estimates are specific to this platform, this algorithm, this campaign, and this time window, and $\tau(M)$ in particular shouldn’t be carried over to a different targeting policy — as we said in Section 5, the algorithm is itself half the reason the bin-level effects look the way they do.

9 Conclusion

Advertising raises the probability of purchase by 0.59 pp (95% CI 0.23-0.95) for the users the algorithm picks, with a net contribution of 0.86 NOK per user (95% CI 0.14-1.58). The effect isn’t a smooth dose-response. It’s close to zero below four impressions, and most of the program’s contribution comes from the small 8-15 bin. We’d recommend keeping the campaign and the current algorithmic targeting, while watching the size of that high-impression group closely, since most of the profitability depends on it.

10 References

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11 Reflection on AI Use

We used AI tools at various points during this assignment, mainly in a supporting role rather than for generating the analysis or writing from scratch.

Tools used: Primarily Google Gemini for code-related tasks, and NotebookLM for revisiting theoretical concepts from the course material.

What we used AI for:

- Debugging code when we ran into errors or unexpected output, and getting suggestions on how to improve code quality (e.g., setting up the manual 2SLS, formulating the delta-method standard error for the Wald LATE, and constructing the binned LATE table).
- Checking for oversights in our analysis (essentially using it as a second pair of eyes to see if we missed something obvious or made a methodological mistake), especially when deciding to add the direct Ghost-Ads ATET as a check against the 2SLS LATE.
- Searching for information to refresh our understanding of the theory behind the methods, particularly around one-sided non-compliance, the LATE/ATET equivalence, and the mechanics of Ghost Ads.
- NotebookLM was useful for going back over course readings and lecture material to make sure we had the concepts right before writing up the discussion, in particular the principal-strata typology and the conditions for the IV exclusion restriction.

How we handled AI input: We treated AI suggestions as starting points rather than final answers. When it flagged something or suggested a change, we checked it against the course material and our own understanding before incorporating it. The decisions to drop the linear interaction, add the direct ATET check, and propagate the LATE CI into the profit calc are our own. The analytical choices, interpretations, and written discussion are our own as well.